

CUSTOMER BEHAVIOUR ANALYSIS

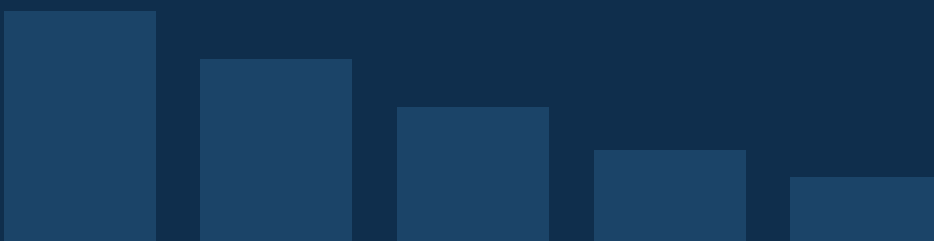
RFM Segmentation & Churn Study
Alfido Tech – Data Analyst Internship

Prepared by

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TOOLS & TECHNOLOGIES

Python	Pandas	NumPy
Matplotlib	Seaborn	Google Colab



1. Project Overview

Understanding how customers buy – how recently, how often and how much – is the foundation of any retention or marketing strategy. This project analyses a large e-commerce transaction dataset to group customers into meaningful segments using RFM analysis, and to examine purchasing patterns across product categories, payment methods, age groups and time.

All figures in this report come directly from the analysis performed in the Google Colab notebook and are limited to the fields available in the dataset.

2. Objective

- Clean and prepare the raw e-commerce transaction data
- Explore purchasing behaviour by product category, payment method and age group
- Track total purchase value over time
- Build an RFM model (Recency, Frequency, Monetary) for every customer
- Group customers into five behavioural segments and profile each one
- Compare churn rates across segments and identify where retention effort should go

3. Dataset

The project uses the **E-commerce Customer Behaviour** dataset (`ecommerce_customer_data_large.csv`), containing **250,000 transactions across 13 columns** and covering purchases from **1 January 2020 to 13 September 2023**. These transactions belong to **49,661 unique customers**.

Column	Description
Customer ID	Unique identifier for each customer
Customer Name, Gender	Customer name and gender
Customer Age / Age	Customer age in years (two identical columns)
Purchase Date	Date and time of the transaction
Product Category	Electronics, Clothing, Home or Books
Product Price	Unit price of the product
Quantity	Number of units bought in the transaction
Total Purchase Amount	Total value of the transaction
Payment Method	Credit Card, PayPal or Cash
Returns	Whether the order was returned (1 / 0, partly missing)
Churn	Whether the customer is flagged as churned (1 / 0)

The dataset does not contain marketing channel, website session, discount or customer location fields, so no claims are made about those areas. Recency, Frequency and Monetary values were derived from the transaction history rather than supplied in the file.

4. Tools & Technologies

Python was used as the core language, with **Pandas** and **NumPy** for cleaning, aggregation and RFM scoring, and **Matplotlib** and **Seaborn** for visualisation. The full analysis was carried out in **Google Colab**.

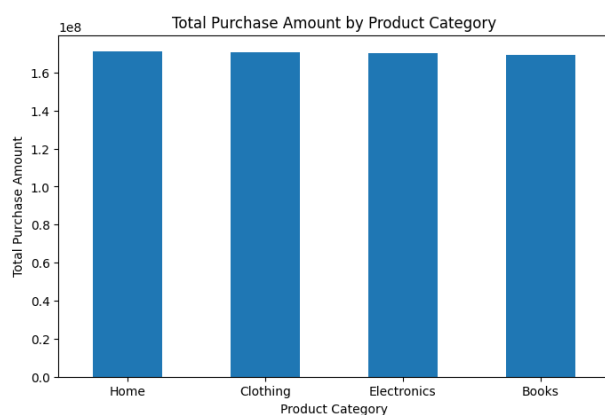
5. Data Cleaning & Preparation

- Loaded the CSV file and inspected structure, data types and summary statistics.
- Checked for duplicate rows – none were found.
- Checked missing values: only **Returns** was affected, with 47,382 blanks. These were spread evenly across categories and payment methods, with near-identical average purchase amounts (2,723.54 vs 2,725.82), so they were treated as unrecorded rather than dropped.
- Created a **Returns Status** column: Returned (101,476), No Return (101,142) or Not Recorded (47,382).
- Converted **Purchase Date** to datetime and derived year, month, month name and day, plus **Age Group** bands (18–25, 26–35, 36–45, 46–55, 56–65, 66+).
- Built a customer-level RFM table using an analysis date of 14 September 2023 (one day after the last purchase).

6. Key Metrics

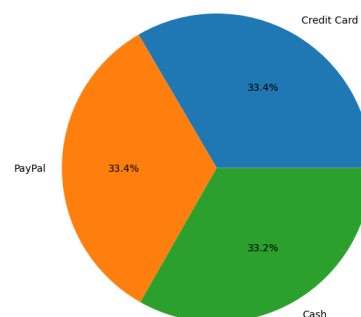
Transactions	250,000	Unique Customers	49,661
Avg Purchase Amount	2,725.39	Avg Recency (days)	262.34
Avg Product Price	254.74	Avg Frequency	5.03
Avg Quantity	3.00	Avg Monetary Value	13,719.95
Avg Customer Age	43.80	Overall Churn Rate	20.05%

7. Purchase Behaviour

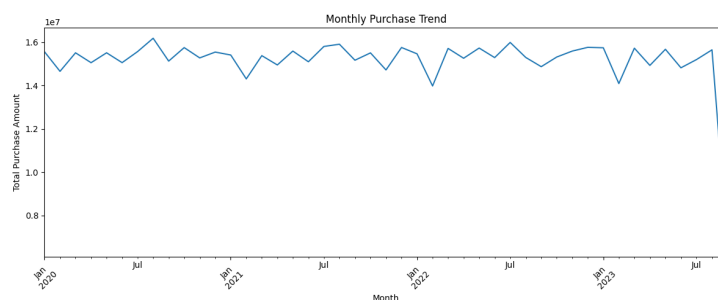


Sales are almost evenly split across the four categories. Home leads with 171.1 million, followed by Clothing (170.7m), Electronics (170.1m) and Books (169.3m) – a spread of under 1%.

Payment Method Distribution



Payment methods are equally balanced: Credit Card 83,547 transactions, PayPal 83,441 and Cash 83,012. No single method dominates.

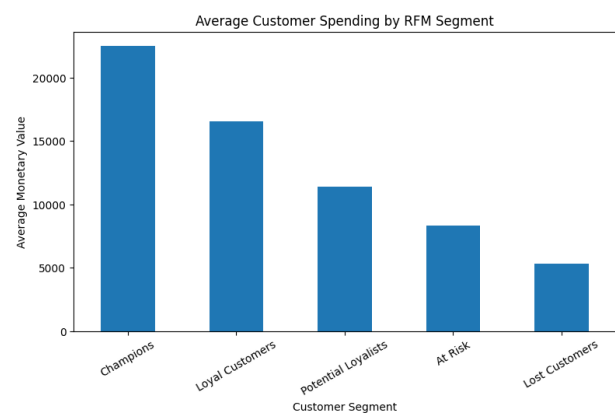
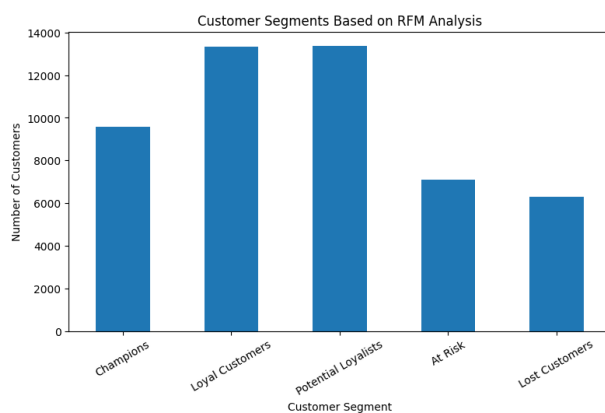


Monthly purchase value is stable at roughly 14–16 million per month, with no growth or seasonal pattern. The fall at the right edge is not real: September 2023 is a partial month.

8. RFM Customer Segmentation

Each of the 49,661 customers was scored from 1 to 5 on Recency, Frequency and Monetary value using quintiles. The three scores were added to give an RFM score between 3 and 15, which was then mapped to five segments: Champions (13–15), Loyal Customers (10–12), Potential Loyalists (7–9), At Risk (5–6) and Lost Customers (below 5).

Segment	Customers	Share	Avg Recency (days)	Avg Frequency	Avg Monetary
Champions	9,573	19.28%	76.73	7.90	22,509.13
Loyal Customers	13,329	26.84%	175.54	6.00	16,574.61
Potential Loyalists	13,367	26.92%	240.03	4.31	11,397.02
At Risk	7,110	14.32%	370.70	3.26	8,307.15
Lost Customers	6,282	12.65%	654.15	2.18	5,338.36



Potential Loyalists (13,367) and Loyal Customers (13,329) are the two largest segments, together making up 53.8% of the customer base. Lost Customers are the smallest group at 6,282.

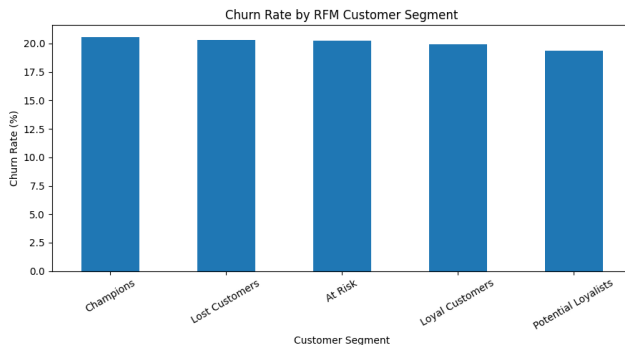
Average spend rises steadily across the segments, from 5,338 for Lost Customers up to 22,509 for Champions – more than four times as much.

Reading the segments

The pattern is exactly what a healthy RFM model should produce. Champions bought most recently (77 days on average), most often (7.9 purchases) and spent the most. Each step down the table shows longer gaps since the last purchase, fewer orders and lower spend, ending with Lost Customers who last bought around 654 days ago and averaged just 2.18 purchases. Champions are 19.28% of customers but carry by far the highest average value, which makes them the priority group to protect.

9. Churn Analysis

The overall churn rate in the dataset is **20.05%**. Churn was then compared across the five RFM segments to see whether behaviour predicts churn.



Churn is essentially flat across every segment. Champions show the highest rate at 20.58%, followed by Lost Customers (20.28%), At Risk (20.24%), Loyal Customers (19.93%) and Potential Loyalists (19.37%).

The entire spread is only 1.21 percentage points, and all five sit close to the 20.05% overall rate, which means the churn flag in this dataset is **not related to purchasing behaviour**. It would be wrong to call Champions the segment most at risk on a difference this small.

10. Key Findings

1. The cleaned dataset covers 250,000 transactions from 49,661 customers between January 2020 and September 2023, with no duplicate records.
2. Purchasing is remarkably even: the four product categories are within 1% of each other on total sales, and the three payment methods within 1% on transaction counts.
3. Monthly purchase value stayed flat at roughly 14–16 million, showing no growth trend or seasonal peak.
4. RFM segmentation separates customers cleanly by value: Champions (19.28%) average 22,509 in spend, over four times the 5,338 averaged by Lost Customers.
5. Churn is flat at around 20% in every segment (19.37% to 20.58%), so the churn flag provides no behavioural signal and cannot currently be used for targeting.

11. Recommendations for Alfido Tech

1. **Protect the Champions segment** – 9,573 customers average 22,509 in spend and bought within the last 77 days. Priority access, loyalty rewards and early product releases defend the most valuable revenue base.
2. **Move Potential Loyalists up a tier** – At 13,367 customers this is the largest segment, averaging 4.31 purchases. Targeted offers and reminder campaigns aimed at raising purchase frequency have the biggest upside.
3. **Run win-back campaigns for At Risk customers** – The 7,110 At Risk customers last bought around 371 days ago but still average 8,307 in spend. Reactivating them is cheaper than acquiring new customers; Lost Customers at 654 days are a lower priority.
4. **Review how churn is defined and recorded** – Because churn sits near 20% in every segment, the current flag carries no behavioural signal. Before any retention model is built, the definition and collection method should be reviewed.
5. **Improve data capture and monitor segments regularly** – Returns were unrecorded for 47,382 transactions (19% of the data), and no channel or location fields exist. Closing these gaps and refreshing the segments monthly would make future analysis far more actionable.

12. Conclusion

This analysis turned 250,000 raw transactions into a clear, customer-level view of behaviour. RFM segmentation works well here, producing five groups that separate sharply by recency, frequency and spend, and giving Alfido Tech a practical basis for deciding which customers to protect, grow or win back.

Equally valuable is what the analysis rules out. Category, payment method and monthly sales are evenly distributed, and the churn flag shows no link to behaviour. Knowing where the signal is not, and saying so plainly, is as much a part of data-driven decision making as the insights.